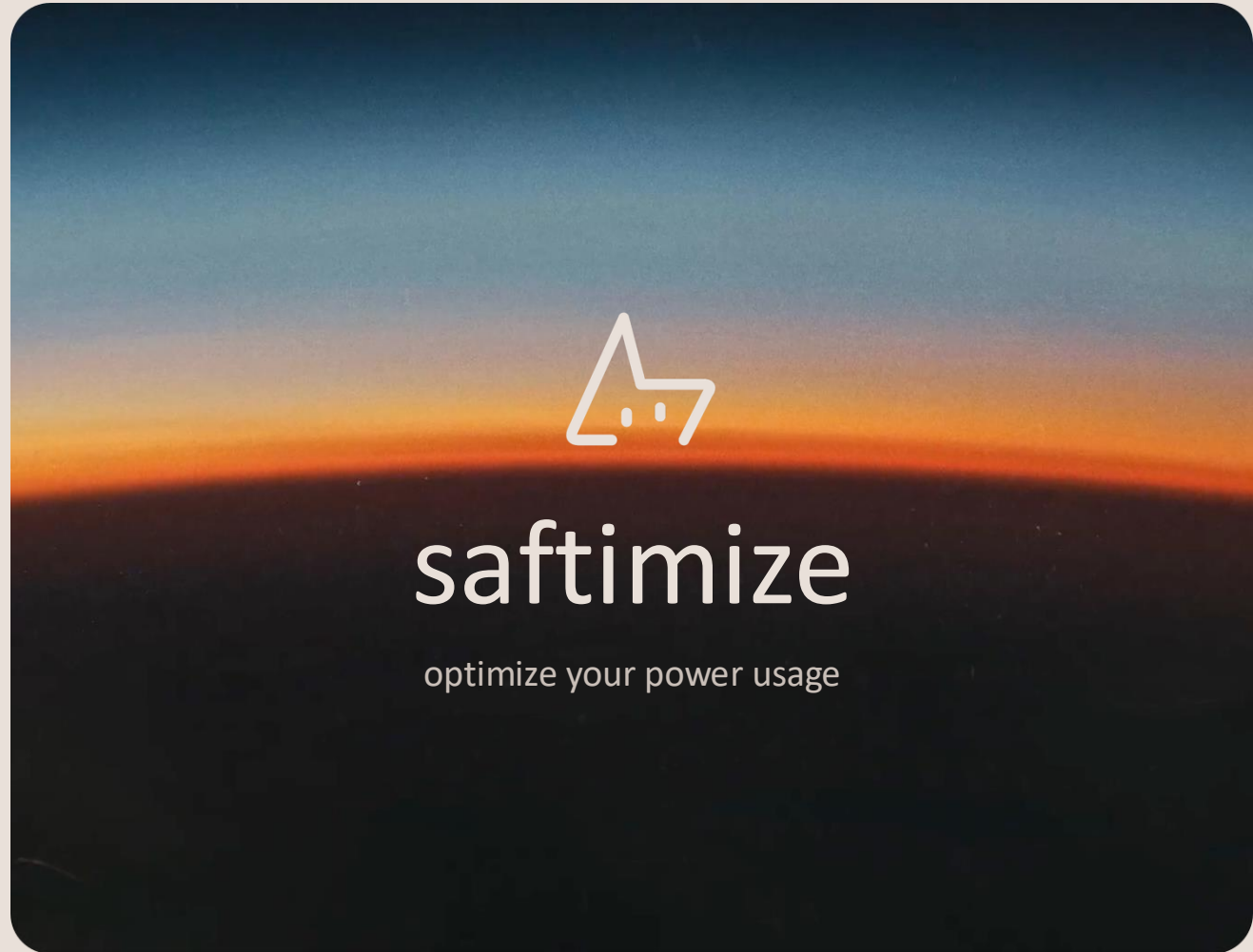


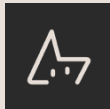


/pitch

Data Engineering HS26
Initial pitch, Timo Schlumpf

See your usage.
Spot the spikes.
Shift the load.





/world

THE COMMON DENOMINATOR

Everything runs on **power.**

Homes, trains, factories, servers.

01

Homes

02

Transport

03

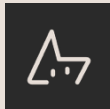
Industry

04

Servers

Take the power away and modern life stops.

The Reason.



/habit

THE BLIND SPOT

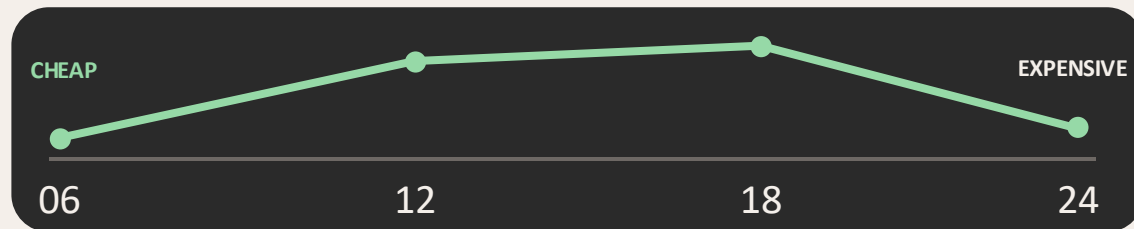
One bill.

per month

WHAT ACTUALLY HAPPENS

Market prices

move hour by hour.



Cheap and expensive hours differ within a single day. Most of us never look.

We rarely plan around it.



/prices

EXTERNAL SHOCKS

01

War and sanctions

The 2022 invasion of Ukraine pushed European gas and power prices to record highs.

02

Weather

Cold snaps, heat waves, low wind and dry rivers shift supply and demand.

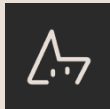
03

Outages

Plant or grid outages can tighten supply within hours.

Switzerland trades power with its neighbours, so these shocks reach Swiss prices too.

The price moves,
always.



/use-case

FROM PERSON → GOAL → PRODUCT

1

Who

Me and my family, at home

2

Goal

See our usage, spot expensive hours ahead, shift flexible loads like laundry

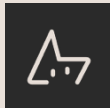
3

Data product

Curated hourly table, dashboard, price forecast and optimization agent (stretch)

HOURLY LOAD + PRICE → DASHBOARD → PRICE FORECAST

The usecase.



/source

THE INPUT LAYER

01

ENTSO-E

Transparency Platform

02

Open-Meteo

Weather Forecast API

DATASET / SCHEMA

ENTSO-E actual load

DateTime(UTC)
ResolutionCode · AreaCode · AreaDisplayName
AreaTypeCode · AreaMapCode
TotalLoad[MW] · UpdateTime(UTC)

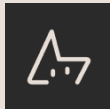
ENTSO-E day-ahead price

InstanceCode · DateTime(UTC) ·
ResolutionCode
AreaCode · ContractType · Sequence
Price[Currency/MWh] · Currency ·
UpdateTime(UTC)

Open-Meteo JSON · hourly.time[] + selected variables[] · hourly_units{} · latitude / longitude / timezone

Daily batch pull · short lookback protects against late / revised source values.

Data Sources.



/architecture

THE PIPELINE

Sources

ENTSO-E + Open-Meteo (Zug)



Ingestion

Python module per source



Raw

Postgres → later GCS



Transform

Clean · align · join · model



Curated

Postgres → later BigQuery

ORCHESTRATOR · daily schedule · retries · backfills

MIDTERM

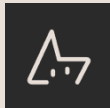
Docker Compose + PostgreSQL

FINAL

Terraform + GCS + BigQuery

Raw stays raw. Transformations create the curated layer.

Architecture V0.1.



/plan

EXECUTION PLAN

Ingestion

Daily incremental load with short
lookback

Idempotent loads, safe to rerun
Retry with backoff, then fail loudly

Storage

Raw data kept unchanged

Postgres locally → GCS + BigQuery for
final

Partition by date, cluster by series

MILESTONES

Pitch

Week 3

Midterm

Repo due 22 Oct

Final

Repo due 10 Dec

MIDTERM

orchestrator + ENTSO-E load/price + raw schema + DST handling

FINAL

Open-Meteo weather + data quality + cloud pipeline

Ingestion,
Storage
& Plan.